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Practical No:1

Applied Multivariate Analysis.

Topic: Descriptive Statistics using R Studio software.

Name: Ramesh Kumar

Roll No: 2K24/STAT/41

Semester: Fifth semester (v)

Instructor: Ms. Hafsa Unar

Submission: 16/05/2026

Objective: - To calculate and analyses the basic descriptive statistics such as mean, median, mode, variance, and standard deviation, and to represent the data graphically using **R Studio**

Introduction: - Descriptive statistics are used to summarize and describe the main features of a dataset. It provides simple numerical measures such as mean, median, and mode to represent the central value of the data, and variance and standard deviation to measure the spread or variability. These measures help in understanding the distribution and behaviours of data. In addition, graphical tools like histograms, bar charts, and pie charts are used to present data visually, making interpretation easier and more effective.

Dataset: The dataset consists of marks obtained by students, ranging from 45 to 95 with a class interval of 5,

Variable: The variable used in this dataset is a quantitative variable, specifically a discrete variable, as it takes specific numerical values, Like

marks of students (45,50,55,60,65,70,75,80,85,90,41)

Total observations: (11)

➤ R Code

Create Variable:

```
marks<-c (45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 41)
```

1.Mean

```
mean(marks)
```

2.Median

```
median(marks)
```

3.mode

```
mode(marks)
```

4.variance

```
var(marks)
```

5.standard deviation

```
sd(marks)
```

6.Minimum

```
min(marks)
```

7.Maximum

```
max(marks)
```

8.Inter Quartile Range

```
IQR (max(marks)-min(marks))
```

9.Range

```
range(marks)
```

10.Summary

```
summary(marks)
```

➤ Output Results from R Studio

```
summary(marks)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	Var.	sd.
41.00	52.50	65.00	65.09	77.50	90.00	270.09	16.43

➤ Interpretation of Results.

Here are the Interpretation Results of given data.

Mean (Average marks of students are 65.09)

Median (50% students scored below 65.00 and 50% scored above. This is the middle value)

Mode (most frequent Repeated value in data, here do not repetition of any values)

1st Quartile (25% of students scored 52.50 or less)

3rd quartile (75% students scored 77.50 or less. While 25% scored above this.)

Maximum (the highest student mark is 90)

Minimum (the lowest student mark is 41)

Variance (this measures the how spread out the marks are from the mean. A larger variance means greater variability)

Standard deviation (on the average deviate from the mean by about 16.43 marks)

Practical No:2

Applied Multivariate Analysis.

Topic: Graphical Representation using R studio

Name: Ramesh Kumar

Roll No: 2K24/STAT/41

Semester: Fifth semester (v)

Instructor: Ms. Hafsa Unar

Submission: 16/05/2026

Objective: - Computation of Graphical Representation using R Studio

Introduction: - Data visualization is an important part of statistics that helps in understanding and presenting data in a clear and meaningful way. Graphs make complex data easier to interpret by showing patterns, trends, and comparisons visually.

a) A bar diagram (bar plot) is used to represent discrete data, where each value is shown as a separate bar. It helps in comparing different categories or individual observations.

b) Histogram: - A histogram is used to display the distribution of continuous data. It groups data into intervals (bins) and shows how frequently values occur within those intervals, helping to understand the shape and spread of the data.

c) Pie chart: - A pie chart represents data as parts of a whole. Each slice shows the proportion or percentage of each category, making it useful for comparing relative sizes. In this case, the population of provinces of Pakistan is shown as percentages to easily understand their share in the total population. ➤ Dataset Description (a) Bar Diagram Data age (18, 19, 12, 13, 15, 21, 22, 23, 25, 19, 13)

d) Line chart: - A line chart is a graphical representation in which data points are plotted and connected by straight lines to show changes or trends over time. It is especially used for time series data, such as GDP, temperature, or population, to display increases, decreases, and fluctuations clearly.

➤ Dataset Description

(a) Bar Diagram Data

age (18, 19, 12, 13, 15, 21, 22, 23, 25, 19, 13)

(b) Histogram Data

height (3, 3.5, 4, 4.5, 5, 5.7, 5.8, 5.9, 6, 6.1, 6.3, 6.5, 5.9)

(c) Pie Chart Data

population pak (127.7, 55.7, 40.9, 14.9)

provinces (Punjab, Sindh, KPK, Baluchistan)

Create Data(description)**Discrete variable:****1. Bar diagram data**

```
Age<-c (18,19,12,13,15,21,22,23,25,19,21)
```

```
Age
```

```
bar plot (Age)
```

```
students<-c("S1","S2","S3","S4","S5","S6","S7","S8","S9","S10","S11")
```

```
bar plot (Age, names.arg = students,
```

```
col = c("red","green","blue","orange","purple","pink","yellow","brown","gray","darkgreen"),
```

```
main = "Age of 11 students" ,xlab="students",ylab="Age")
```

Continuous variable:**2. Histogram data**

```
Height<-c (3,3.5,4,4.5,5,5.7,5.8,5.9,6,6.1,6.3,6.5,5.5)
```

```
Height
```

```
hist (Height)
```

```
students<-c("S1","S2","S3","S4","S5","S6","S7","S8","S9","S10","S11","S12","S13")
```

```
hist (Height, names.arg=students,
```

```
col=c("red","green","blue","orange","purple","pink","yellow","cyan","brown","gray","darkgreen","gold",  
"lightblue"),
```

```
main = "Height of 13 Students", xlab="Height (cm)", ylab= "Students")
```

3. Pie chart data

```
provinces<-c ("Sindh", "Punjab", "KPK", "Baluchistan")
```

```
provinces
```

```
papulation<-c (55.7,127.7,40.9,14.9)
```

```
papulation
```

```
pie (papulation, labels = provinces, main = papulation)
```

```
percent<-round(papulation/sum(papulation)*100)
```

```
labels<-paste (provinces, percent, "%")
```

```
pie (papulation, labels = labels, main = "papulation distribution of Pakistan province's (2023)",
```

```
col = c("green","purple","orange","red"), radius = 1.0)
```

```
pie (papulation, lables=lables, col = c("green","purple","orange","red"), main = "papulation  
distribution")
```

```
legend ("topleft", legend = provinces, fill = c("green","purple","orange","red"), cex = 0.6, pt.cex =  
0.5)
```

4. Line chart data

```
year <- c (2000,2001,2002,2003,2004,2005,2006,2007,2008,2009,2010)
```

```
gdp <- c (99.48,97.15,97.92,112.37,132.22,145.27,161.87,184.14,202.2,187.34,196.71)
```

```
plot (gdp, type = "o", col="blue", lwd=2, pch=16, main = "GDP of pakistan (2000-2010)", xlab =  
"Year", ylab = "GDP (Billions USD)",
```

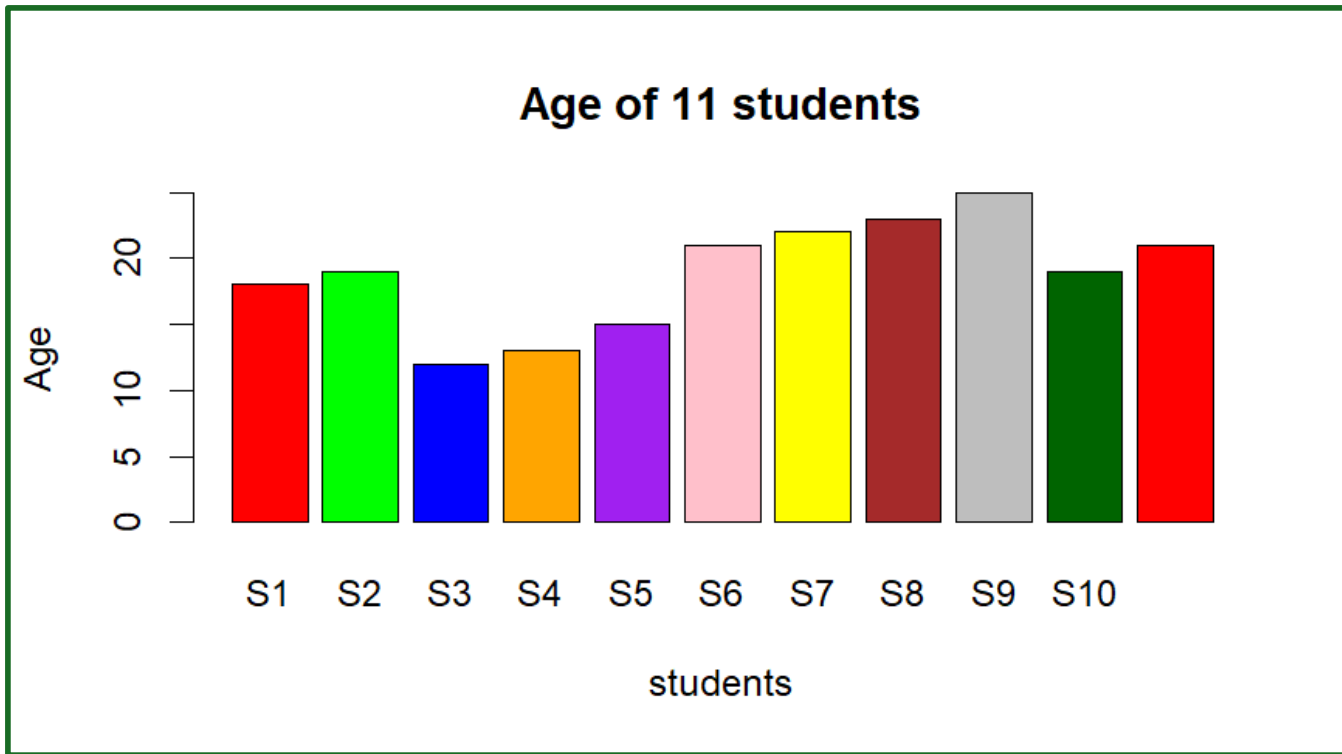
```
lines (year, gdp, col="blue", lwd=2)
```

```
points (year, gdp, pch=16, col="blue")
```

```
plot (year,gdp,type = "o")
```

➤ Graphical Representation Output from R studio.

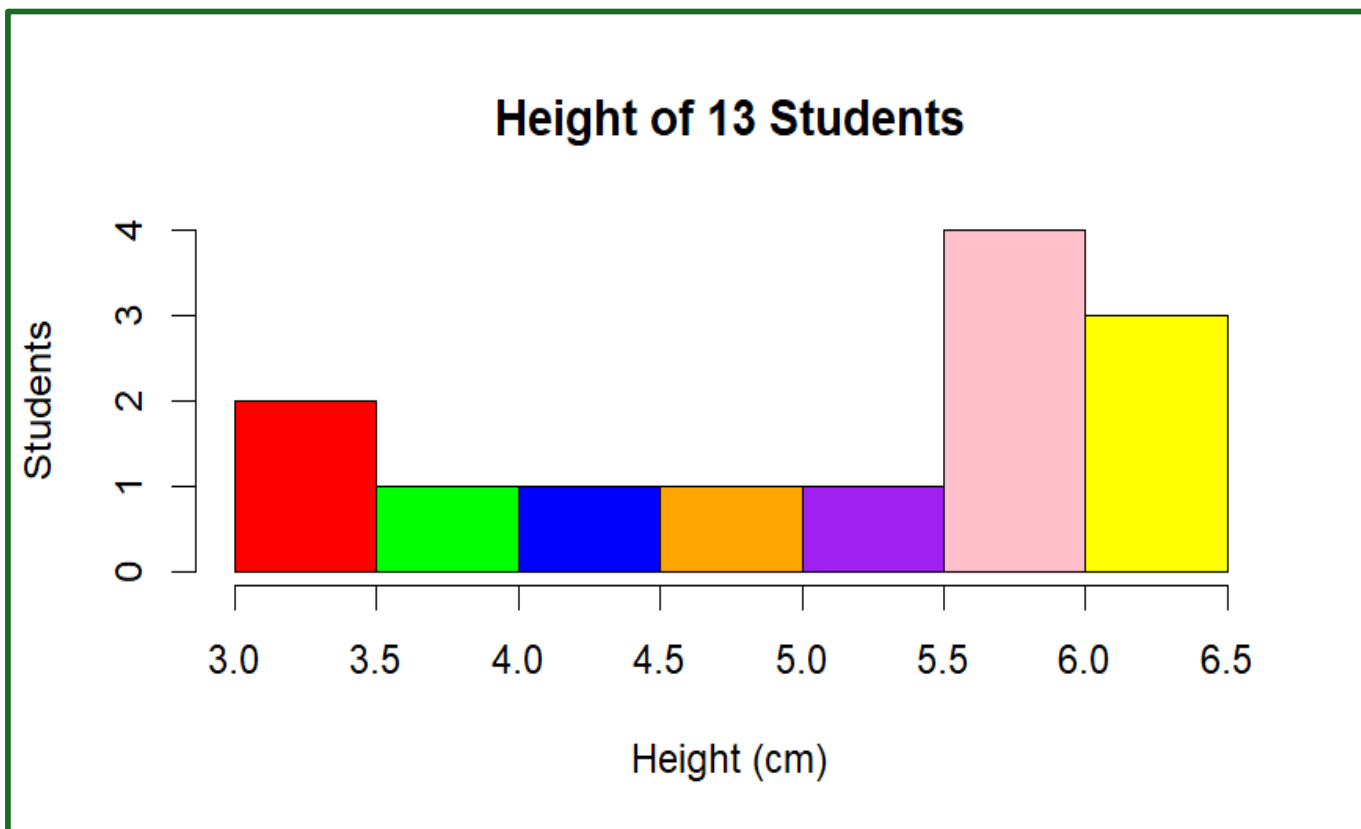
1. Bar Diagram/plot of Students Age.



1. Bar Plot Interpretation (Age of Students)

The bar plot shows the age distribution of 11 students. Ages range from 12 years to 25 years, indicating variation in students' ages. Most students fall in the late-teen to early-twenties range, with ages 19 and 21 appearing more than once, suggesting these are common ages in the group. The graph helps compare the age of each student individually.

2. Histogram of Students Height.

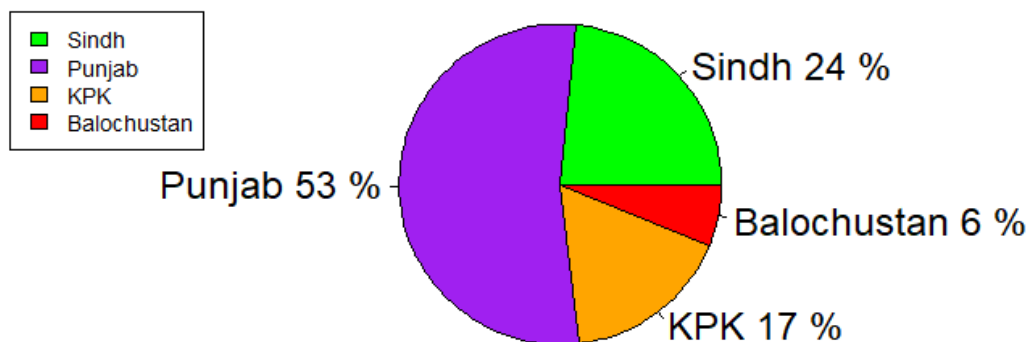


2. Histogram Interpretation (Height of Students)

The histogram shows the distribution of heights for 13 students. Heights range from 3 feet to 6.5 feet. Most observations appear concentrated around 5 to 6.5 feet, indicating that many students have heights in this interval. Very low heights (such as 3 and 3.5 feet) occur less frequently. The histogram helps show the spread and frequency pattern of the height data.

3. Pie chart of Pakistan Provinces Papulation Distribution.

population distribution of pakistan provoinces(2023)

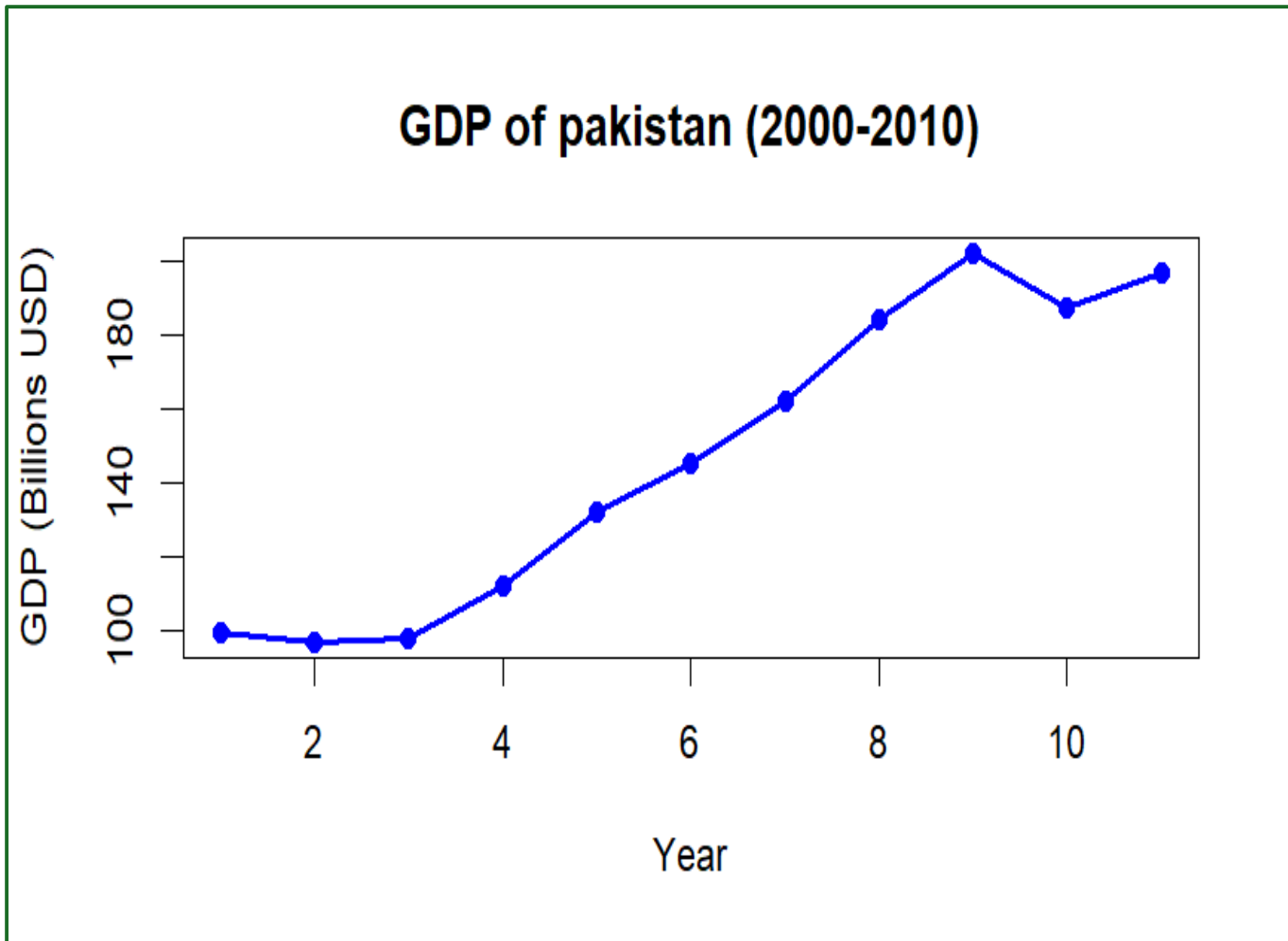


3. Pie Chart Interpretation (Population of Pakistan Provinces)

The pie chart shows the population distribution among provinces of Pakistan. Punjab has the largest share of the population (127.7), followed by Sindh (55.7), KPK (40.9), while Baluchistan (14.9) has the smallest share. This indicates that population is concentrated mainly in Punjab, whereas Baluchistan has comparatively low population.

Punjab = 127.7 million

4. Line chart GDP of Pakistan.



4. Line chart Interpretation (GDP of Pakistan)

The line chart represents the GDP of Pakistan from 2000 to 2010. The graph shows that GDP generally followed an increasing trend over the study period, rising from 99.48 billion in 2000 to 196.71 billion in 2010. GDP increased steadily up to 2008, reaching its highest value of 202.20 billion, followed by a decline in 2009. A recovery is observed in 2010. Overall, the line chart indicates long-term economic growth with a temporary fluctuation.

Practical No: 3

Applied Multivariate Analysis.

Topic: Matrix and its types.

Name: Ramesh Kumar
Roll No: 2K24/STAT/41
Semester: Fifth semester (v)
Instructor: Ms. Hafsa Unar
Submission: 16/05/2026

Objective: -To study the concept of matrices and their dimensions.

to understand transpose of a matrix and different types of matrices.

to identify row, column, square, diagonal, scalar, identity, symmetric, and skew-symmetric matrices.

Introduction: -Matrix is an arrangement of numbers or elements in rows and columns enclosed in brackets. Matrices are widely used in statistics, algebra, economics, and computer science for solving systems of equations, data organization, and mathematical analysis.

The dimension (order) of a matrix is written as $m \times n$, where m is the number of rows and n is the number of columns. The transpose of a matrix is obtained by interchanging rows into columns and is denoted by T .

➤ Dataset Description:

1. Row Matrix

A matrix having only one row and any number of columns is called a row matrix.

Its order is 1×3

Example: (4, 7, 10)

2. Column Matrix

A matrix having only one column and any number of rows is called a column matrix.

Its order is 3×1 .

Example: (4, 7, 10)

3. Square Matrix

A matrix having equal numbers of rows and columns is called a square matrix.

Its order is 4×4

Example: (4, 7, 3, 9)

4. Diagonal Matrix

A square matrix in which all non-diagonal elements are zero is called a diagonal matrix.

Only principal diagonal elements may be non-zero.

Example: (52,0,0,0,0,23,0,0,0,0,49,0,0,0,0,45)

5. Scalar Matrix

A diagonal matrix whose diagonal elements are equal is called a scalar matrix.

All off-diagonal elements are zero.

Example: (2,0,0,0,2,0,0,0,2)

6. Identity Matrix

A scalar matrix whose diagonal elements are all 1 is called an identity matrix.

It is denoted by I .

Example: (1,0,0,0,1,0,0,0,1)

7. Symmetric Matrix

A square matrix is symmetrical if its transpose equals itself, i.e.

$T = S$

Example: (2,2,17,2,5, -11,17, -11,9)

➤ R code

1.Row matrix

```
R<-matrix(c(4,7,10), nrow = 1)
```

```
R
```

2.Column matrix

```
v<- matrix(c(4,7,10), ncol = 1)
```

```
v
```

3. Square matrix

```
S<-matrix(c(4,7,3,9), nrow = 2, byrow = TRUE)
```

```
S
```

```
det(S)
```

```
t(S)
```

4. Diagonal matrix

```
D<-matrix(c (52,0,0,0,0,23,0,0,0,0,0,49,0,0,0,0,45), row = 4, byrow = TRUE)
```

```
D
```

```
t(D)
```

```
det(D)
```

5. Scalar matrix

```
s<-matrix(c(2,0,0,0,2,0,0,0,2), nrow = 3, byrow = TRUE)
```

```
s
```

```
t(s)
```

```
det(s)
```

6. Identity matrix

```
I<-matrix(c(1,0,0,0,1,0,0,0,1), nrow = 3, byrow = TRUE)
```

```
I
```

```
t(I)
```

```
det(I)
```

7.Symmetric matrix

```
T<-matrix(c(2,5,7,10), nrow = 2, byrow = TRUE)
```

```
T
```

```
t(T)
```

```
det(T)
```

➤ Output Result from R studio

1. row matrix

```
4  7  10
```

2. column matrix

```
4
7
10
```

3. Square matrix

```
4  7
3  9
```

t(s)

```
4  3
7  9
```

det(s)

```
15
```

4. Diagonal matrix

```
52  0  0  0
0  23  0  0
0  0  49  0
0  0  0  45
```

t(D)

```
52  0  0  0
0  23  0  0
0  0  49  0
0  0  0  45
```

det(D)

```
2637180
```

5. Scalar matrix

```
2  0  0
0  2  0
0  0  2
```

t(s)

```
2  0  0
0  2  0
0  0  2
```

det(s)

```
8
```

6. Identity matrix

```
1  0  0
0  1  0
0  0  1
```

t(I)

```
1  0  0
0  1  0
0  0  1
```

det(I)

```
1
```

7. Symmetric matrix

$$\begin{pmatrix} 2 & 5 \\ 7 & 10 \end{pmatrix}$$
$$t(T) = \begin{pmatrix} 2 & 5 \\ 7 & 10 \end{pmatrix}$$
$$\det(T) = -15$$

➤ Interpretation of the results

The practical demonstrates that matrices can be classified according to their order and special properties. A row matrix contains only one row, while a column matrix contains only one column. A square matrix has equal rows and columns, which allows special forms such as diagonal, scalar, and identity matrices.

It is observed that diagonal matrices have non-zero elements only on the principal diagonal, scalar matrices have equal diagonal elements, and identity matrices have all diagonal elements equal to one. Symmetric matrices satisfy $A = A'$ while skew-symmetric matrices satisfy $A = -A'$. Hence, the practical helps in understanding dimensions, transpose, and the properties of different types of matrices.

Practical No:4

Applied Multivariate Analysis.

Topic: Linear model through matrix.

Name: Ramesh Kumar

Roll No: 2K24/STAT/41

Semester: Fifth semester (v)

Instructor: Ms. Hafsa Unar

Submission: 16/05/2026

Objective: - Computation of Linear Model Trough Matrices Using R Studio.

➤ **Introduction (Theory):** - First, the data was entered into R Studio in matrix form. A design matrix X was created by adding a column of ones along with the variable X. Then, the transpose of X was taken. After that, the matrix $X'X$ was calculated and its inverse was found. Next, the matrix $X'Y$ was computed by multiplying the transpose of X with Y. Finally, the regression coefficients were obtained by multiplying the inverse of $X'X$ with $X'Y$.

➤ Dataset Description:

Lot Size (X): 30, 20, 60, 80, 40, 50, 60, 30, 70, 60, 41

Man Hours (Y): 41, 73, 50, 128, 170, 87, 108, 135, 69, 148, 132

Number of observations: 11

➤ R code

```
x<-matrix(c(30,20,60,80,40,50,60,30,70,60,41), nrow = 11, ncol = 1)
x
y<-matrix(c(41,73,50,128,170,87,108,135,69,148,132), nrow = 11, ncol = 1)
y
```

1. Design matrix

```
x<-cbind(1, x)
x
```

2. X'

```
xt<-t(x)
xt
```

3. $X'X$

```
xtx<-xt%*%x
xtx
```

4. $X'Y$

```
xty<-xt%*%y
xty
```

5. $(X'X)^{-1}$

```
inv<-solve(xtx)
inv
```

6. Y

```
y<-matrix(y,ncol = 1)
y
```

7. β

```
beta<-solve(t(x)%*%x)%*%t(x)%*%y
beta
```

8. Model

```
model<-lm(y~x)
model
```

9. Residuals

```
yhat<-x%*%beta
yhat
res<-y-yhat
res
```

10. Residuals plot

```
plot(yhat, res, col="darkgreen", pch=19, main = "residual plot", xlab = "Fitted values", ylab = "Residuals")
abline(h=0, col="red", lwd=2)
```

11. Scatter Plot

```
plot(x,y, col="blue", pch=19, main = "Scatter Plot", xlab = "Lot size", ylab = "Man-hours")  
abline(model, col="red", lwd=2)
```

➤ Output result from R studio

1. X

$$\begin{pmatrix} 30 \\ 20 \\ 60 \\ 80 \\ 40 \\ 50 \\ 60 \\ 30 \\ 70 \\ 60 \\ 41 \end{pmatrix}$$

2. Y

$$\begin{pmatrix} 41 \\ 73 \\ 50 \\ 128 \\ 170 \\ 87 \\ 108 \\ 135 \\ 69 \\ 148 \\ 132 \end{pmatrix}$$

3. Design matrix

$$\begin{pmatrix} 1 & 30 \\ 1 & 20 \\ 1 & 60 \\ 1 & 80 \\ 1 & 40 \\ 1 & 50 \\ 1 & 60 \\ 1 & 30 \\ 1 & 70 \\ 1 & 60 \\ 1 & 41 \end{pmatrix}$$

4. X'

1	1	1	1	1	1	1	1	1	1	1	1	1	1
30	20	60	80	40	50	60	30	70	60	41			

5. X'X

11	541
541	30081

6. (X'X)⁻¹

0.7872546	-0.0141585972
-0.0141586	0.000287882

7. X'Y

1141
56732

8. beta

95.0120126
0.1772049

9. Estimated model/ Regression Equation

$$(\hat{Y} = 95.89 + 0.1772X)$$

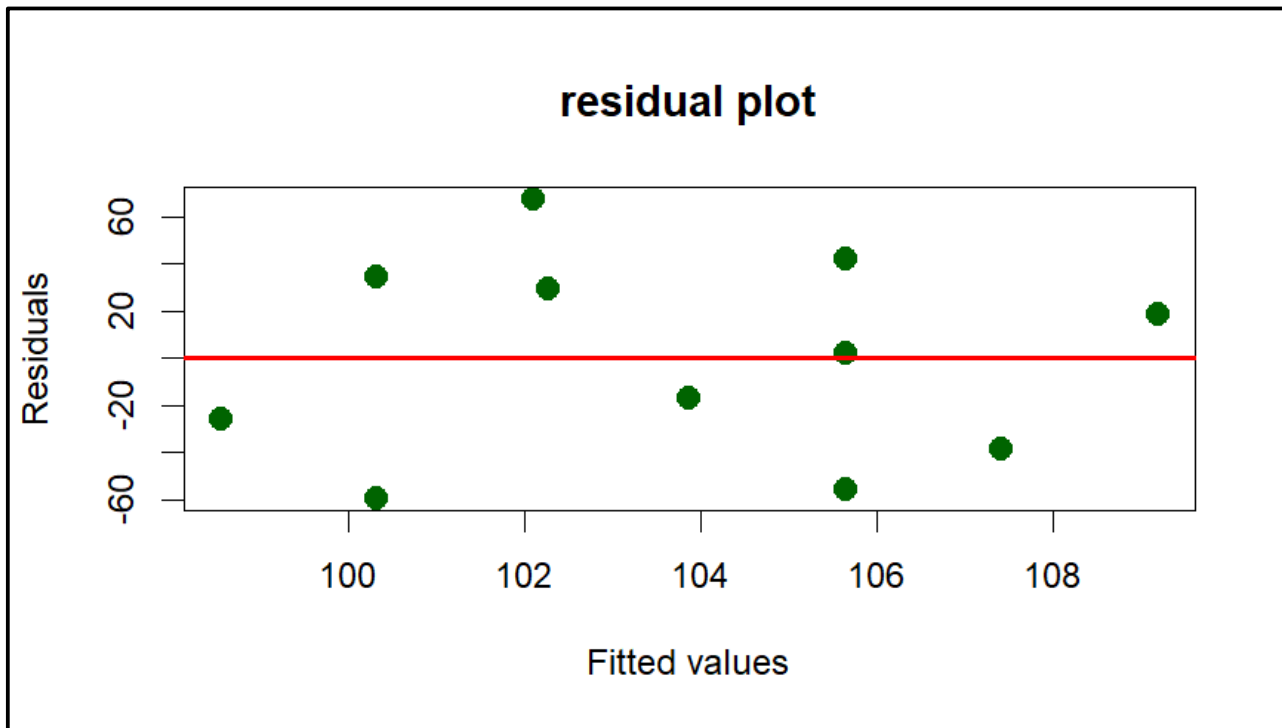
10. Y hat

100.32816
98.55611
105.64431
109.18841
102.10021
103.87226
105.64431
100.32816
107.41636
105.64431
102.27741

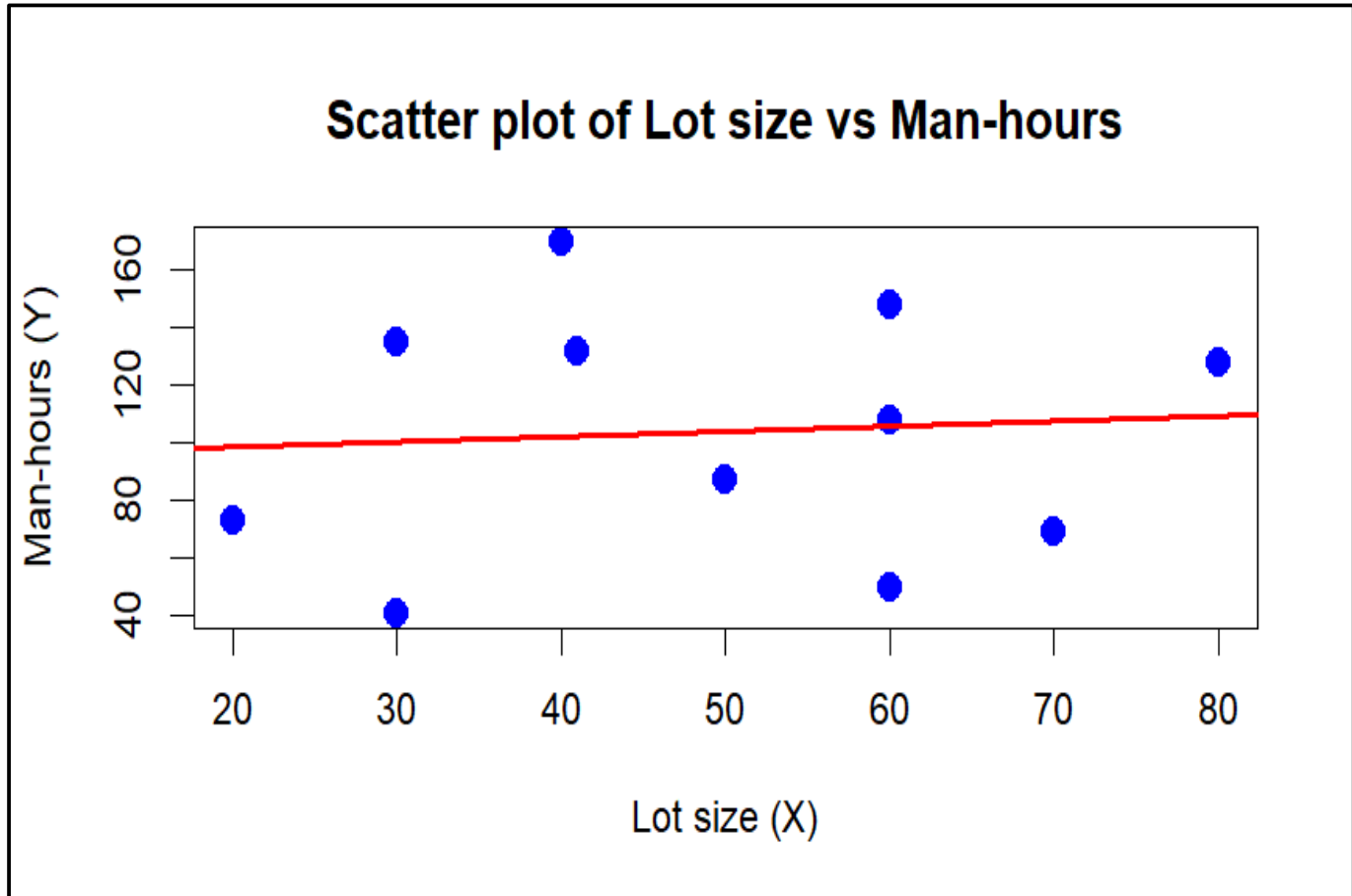
11. Residuals

59.328160
-25.556111
-55.644308
18.811594
67.899791
-16.872259
2.355692
34.671840
-38.416357
42.355692
29.722586

- **Graph:**
- **Residuals Plot: -**



- **Scatter Plot: -**



➤ Interpretation of Results: -

The regression equation ($Y = 95.8959 + 0.1772X$) shows the relationship between Lot Size (X) and Man Hours (Y). The value 95.8959 is the starting point of man hours when lot size is zero. The value 0.1466 means that when lot size increases by 1 unit, man hours also increase slightly. This shows a positive relationship between both variables. The scatter plot shows that as lot size increases, man hours also increase in an upward direction. The regression line fits the data well. The residual plot shows random points around zero, which means the model is good and no major pattern is left. Overall, there is a clear positive linear relationship between lot size and man hours.

Practical No:5

Applied Multivariate Analysis.

Topic: Create Random vector.

Name: Ramesh Kumar

Roll No: 2K24/STAT/41

Semester: Fifth semester (v)

Instructor: Ms. Hafsa Unar

Submission: 16/05/2026

Objectives: - To construct a random vector from bivariate data.
to calculate the expected value (mean vector).
to find the variance-covariance matrix.
to study variability and relationship between two variables.

Introduction: -

A random vector is a vector whose elements are random variables. If two random variables X and Y are studied together, they can be represented as:

The expected value gives the average behavior of the variables, while the variance-covariance matrix measures variability and association between them.

Data Description: -

(X): 44, 48, 49, 41.

(Y): 45, 52, 23, 41.

➤ R code

1. Create random vector data

```
x<-c(44,48,49,41)
x
y<-c(45,52,23,41)
y
```

2. Combine as random vector (data matrix)

```
z<-cbind(x,y)
```

3. mean vector expected value

```
colMeans(z)
```

4. variance covariance matrix

```
cov(z)
```

5. Individual variances

```
var(x)
var(y)
```

6. covariance only

```
cov(x,y)
```

➤ Output Result from R studio

1. x

```
44 48 49 41
```

2. y

```
45 52 23 41
```

3. Expected value/means vector.

$$\begin{pmatrix} x & y \\ 45.50 & 40.25 \end{pmatrix}$$

4. variance co-variance matrix.

$$\begin{pmatrix} x & y \\ x & 13.66667 & -13.83333 \\ y & -13.83333 & 152.91667 \end{pmatrix}$$

5. variances of x and y.

```
x
13.66667
y
152.9
```

6. co-variance of x and y.

```
-13.83333
```

➤ **Interpretation of results**

“The mean vector shows average values of X and Y are 45.5 and 40.25 respectively. The covariance matrix shows X has low variability, while Y has high variability. The negative covariance (-13.83) indicates an inverse association between X and Y, meaning they tend to move in opposite directions.”